

CMPE2600 Lab 3 – Chem Plant – Displays Levels Using HS Counter and DP Transducer

Name: _____

Introduction

This project has multiple phases. All phases relate to designing and implementing a PLC program to control a chemical process. The actual plant uses 4 tanks: 2 chemical storage tanks, a mixing tank to heat and stir the two chemicals before dumping them to a holding tank. You will be programming the controller for a simulator that has a mixing tank and a reservoir representing the source tanks for the two chemicals and the holding tank.

System Description (effectively the Problem Statement)

If the Start button is pressed the system enters Normal mode which operates as follows:

- In this lab we want to display the height in CM when filling using high speed counter. You will need to set up a high-speed counter just like in lab 20.(Proving the turbine meter)
- Also display the height in engineering units when the tank is draining. You will need to use the output from the Differential Pressure cell which is connected to ADC3. Design the transfer function needed to appropriately convert to engineering units using the span of the Analog Input. Remember to account for the 5cm offset at the bottom as 0 cm = 0 EU and 60cm is 999 EU.
- All this needs to be displayed on the main message.

Check off _____